

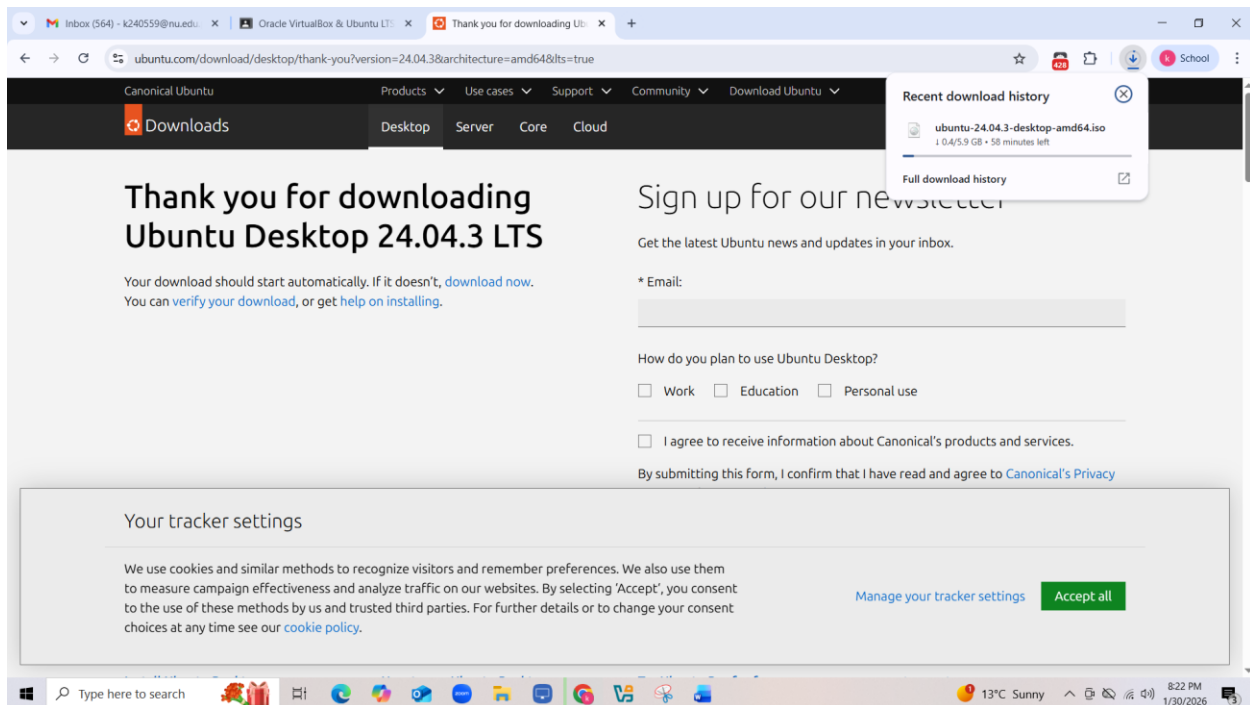
24k-0559

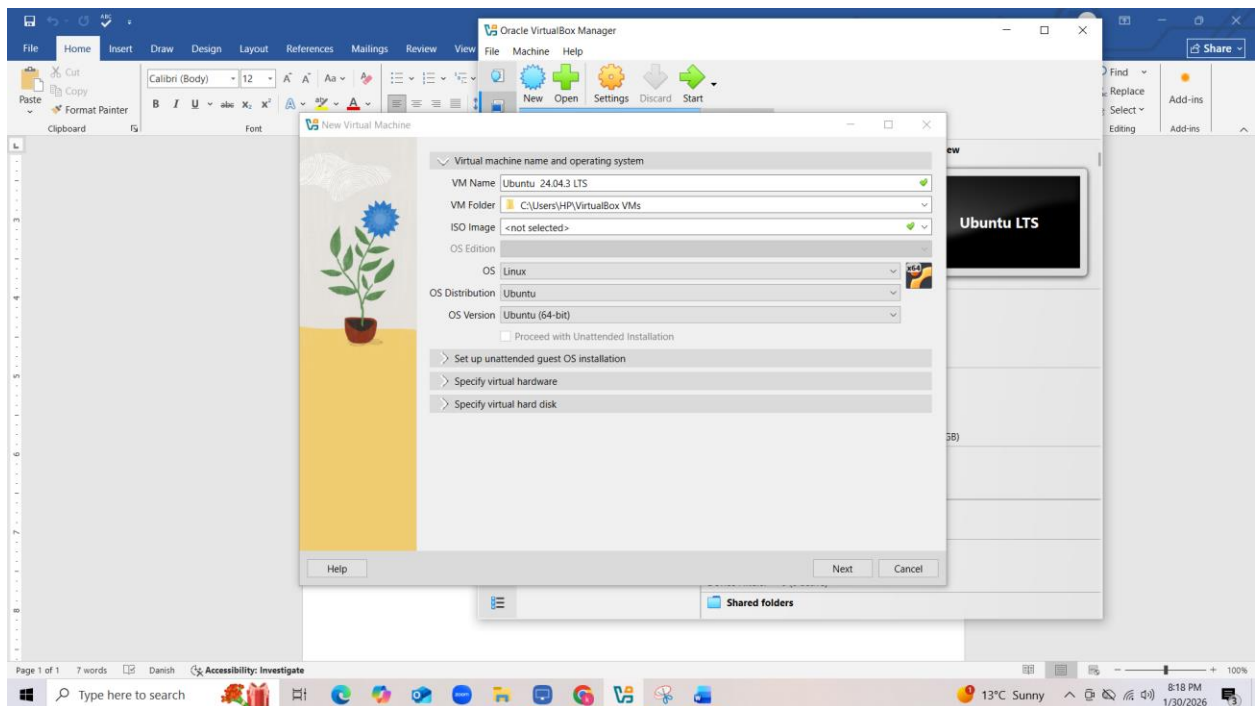
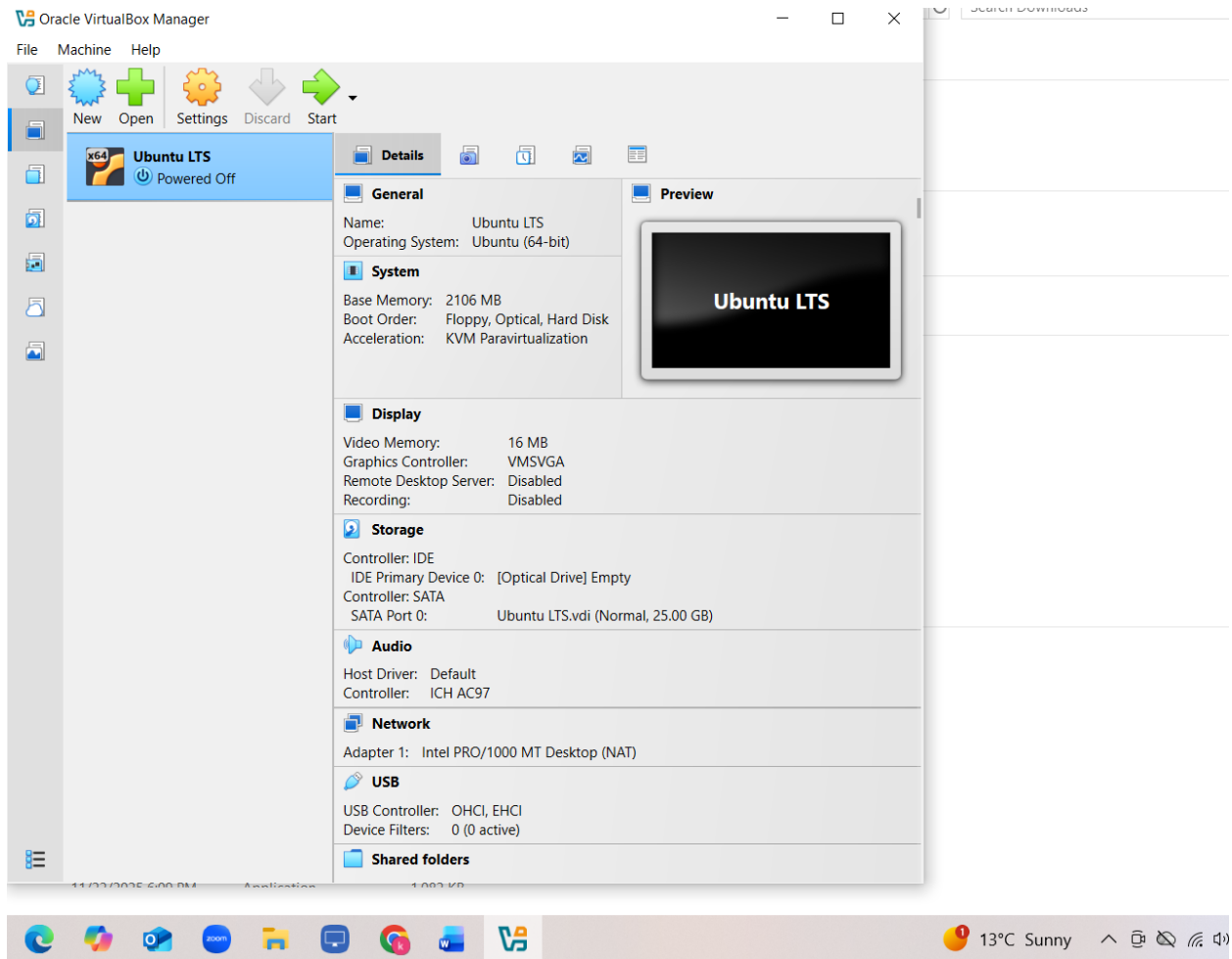
BAZIL-UDDIN-KHAN

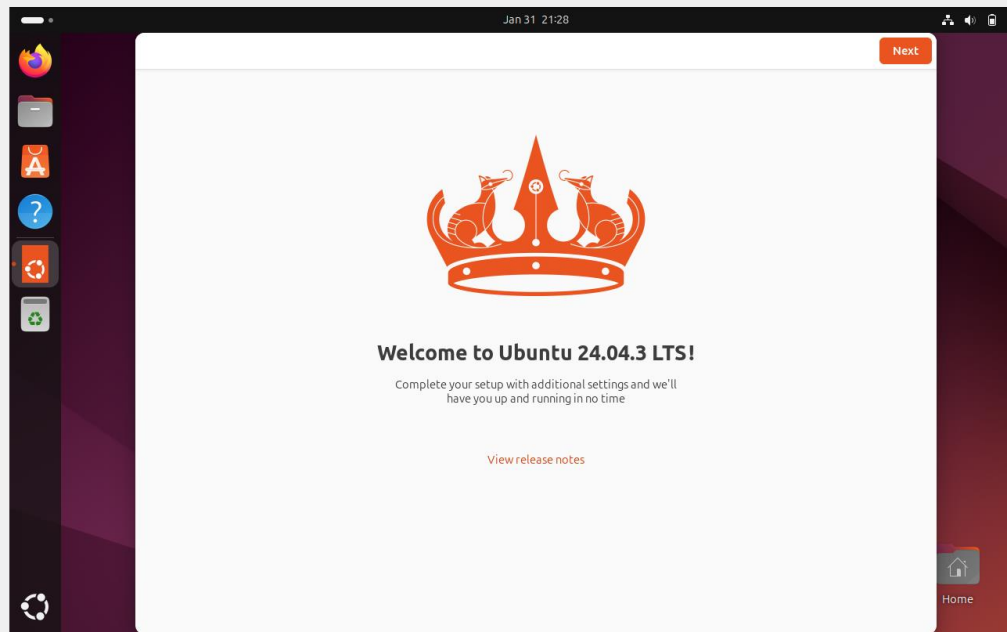
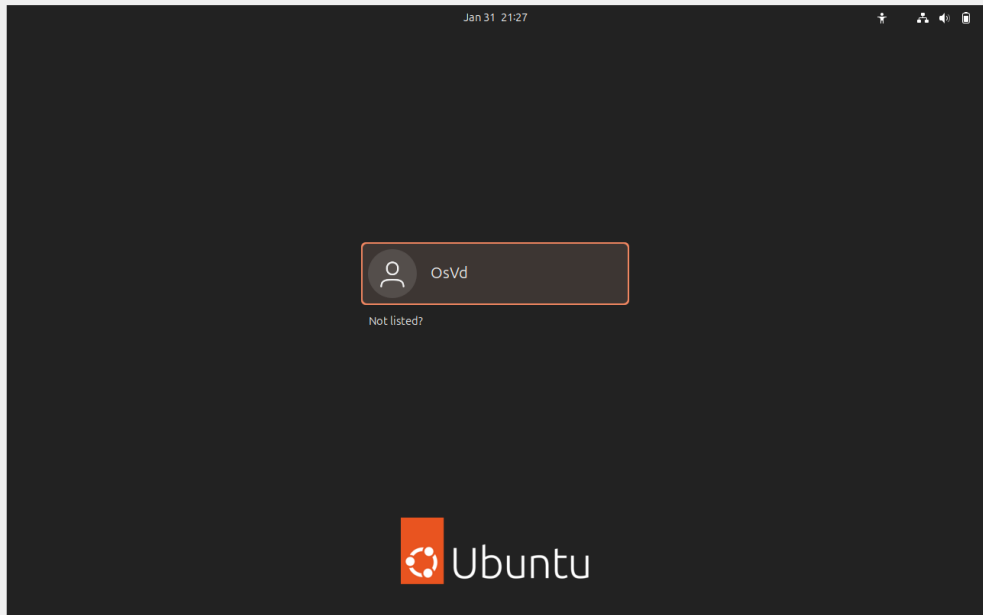
BSCS 4H

OPERATING SYSTEMS:

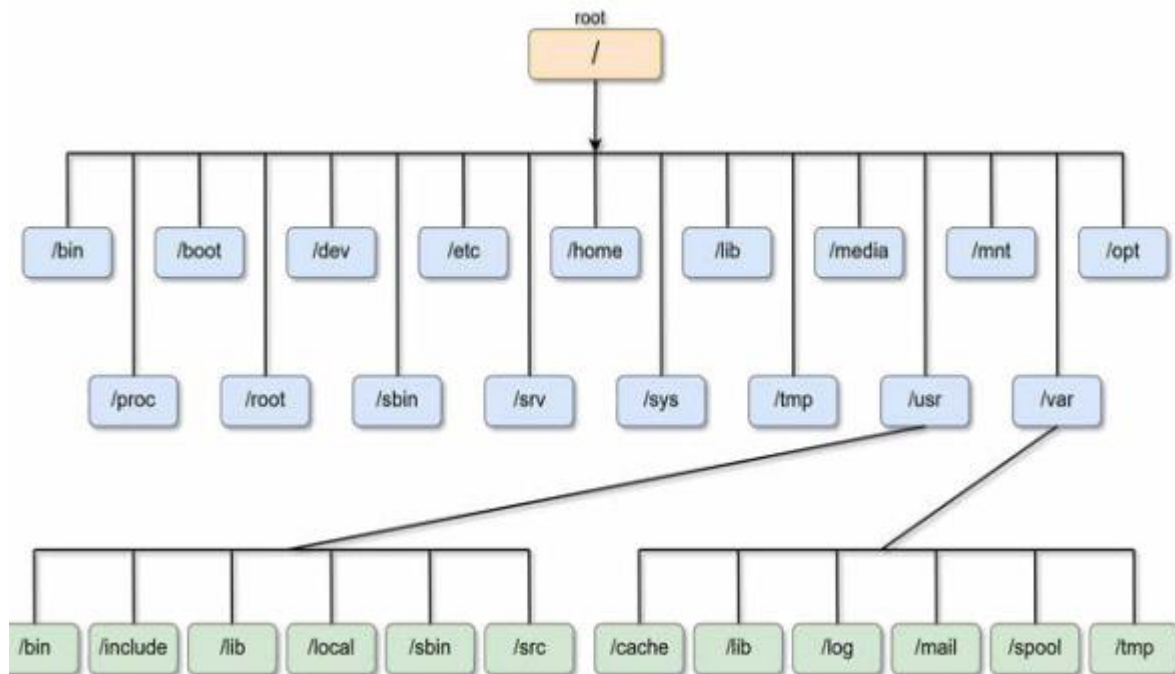
TASK#1).







TASK#2).

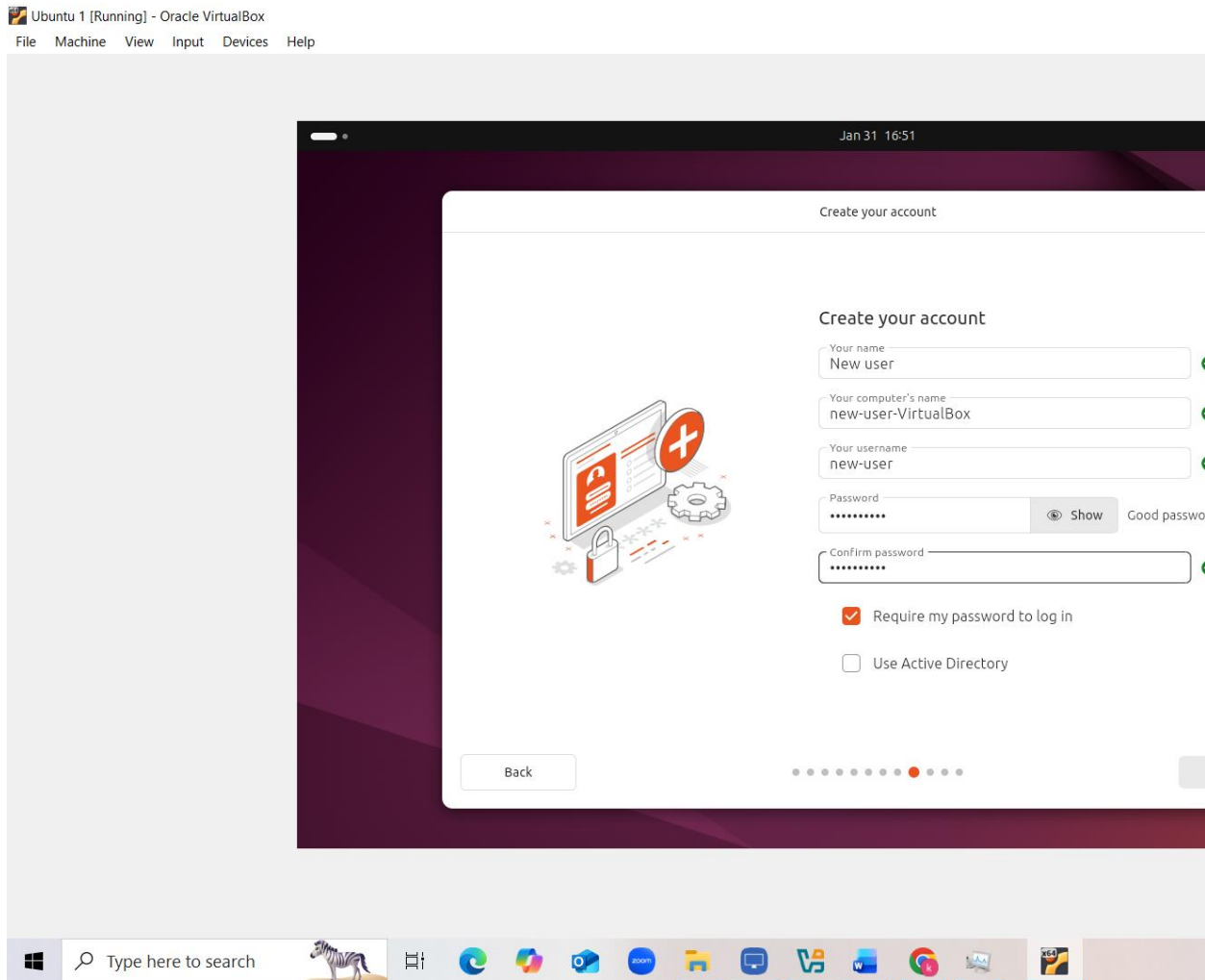


Linux File System

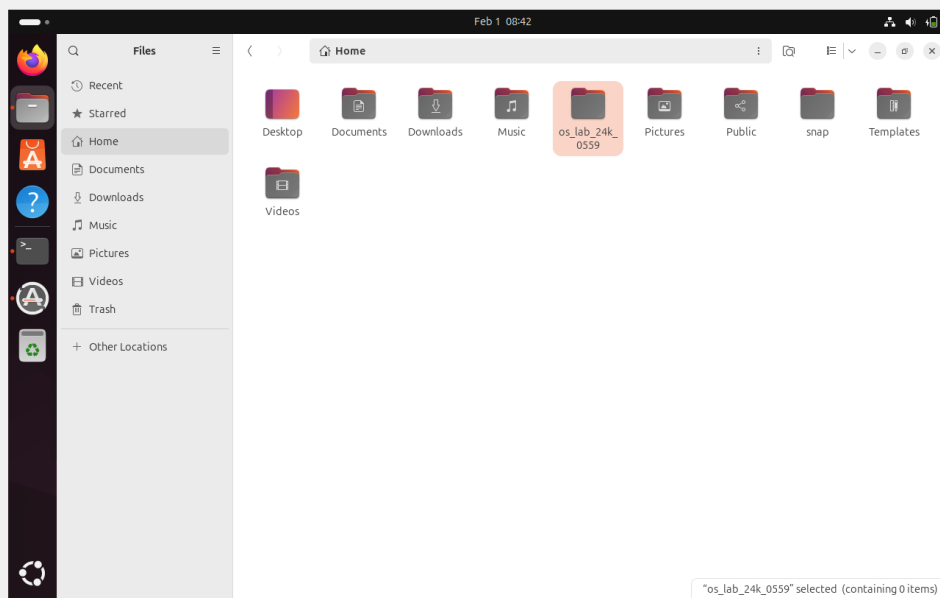
1. **/:** Root and it is the top of the file system tree . Every file and directory on the system exists under / and known as parent.
2. **/ bin:** it Contains essential user commands. These commands are needed for the system to work in single-user mode .
3. **/sbin:** Contains system administration commands and Mainly used by the root user .
4. **/etc:** In this whole text files are contained. It don't have executable programs.
5. **/home:** This directory will contain all the personal home directories for all regular users on the system .
6. **/root :** This is the home directory for the root user (the system administrator).
7. **/usr :** It contains read-only data for applications.
8. **/var :** This directory contains files that can change in size or content frequently.
9. **/tmp :** This directory is used for storing temporary files created by the system or applications.

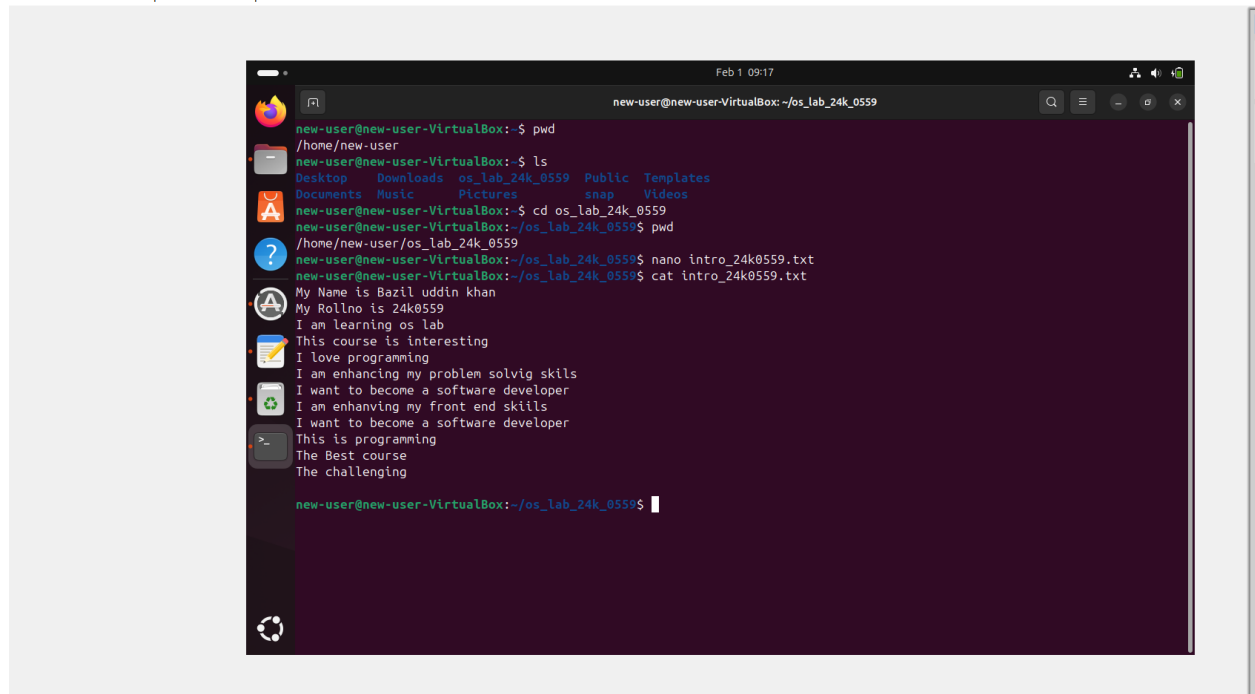
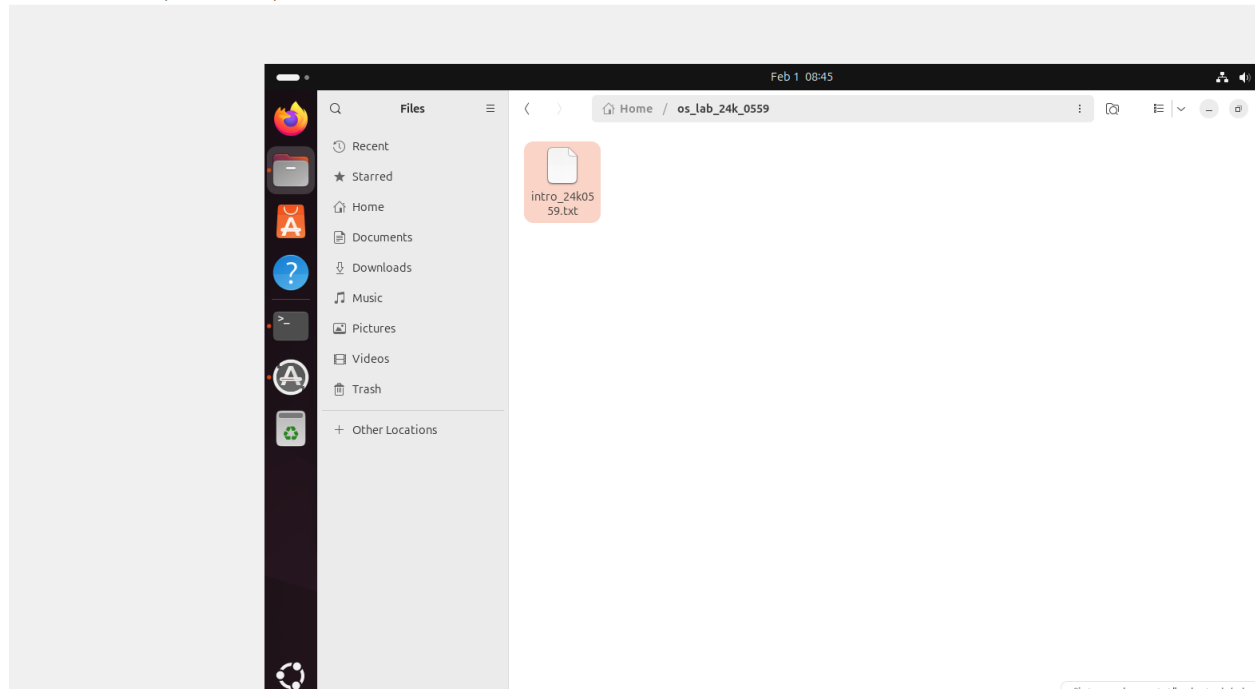
- 10. **/dev** : This directory contains special file representations of hardware devices like usb drive etc.
- 11. **/proc** : A virtual file system that does not exist on the disk, but in memory
- 12. **/lib** : This contains shared library images needed to boot the system and run commands in the root file system.

TASK#3).



```
new-user@new-user-VirtualBox: ~  
Setting up libctf0:amd64 (2.42-4ubuntu2.8) ...  
Setting up libgprofng0:amd64 (2.42-4ubuntu2.8) ...  
Setting up libgcc-13-dev:amd64 (13.3.0-6ubuntu2-24.04) ...  
Setting up libstdc++-13-dev:amd64 (13.3.0-6ubuntu2-24.04) ...  
Setting up binutils-x86-64-linux-gnu (2.42-4ubuntu2.8) ...  
Setting up gcc-13-x86-64-linux-gnu (13.3.0-6ubuntu2-24.04) ...  
Setting up binutils (2.42-4ubuntu2.8) ...  
Setting up dpkg-dev (1.22.6ubuntu6.5) ...  
Setting up gcc-13 (13.3.0-6ubuntu2-24.04) ...  
Setting up g++-13-x86-64-linux-gnu (13.3.0-6ubuntu2-24.04) ...  
Setting up gcc-x86-64-linux-gnu (4:13.2.0-7ubuntu1) ...  
Setting up gcc (4:13.2.0-7ubuntu1) ...  
Setting up dkms (3.0.11-1ubuntu13) ...  
Setting up g++-x86-64-linux-gnu (4:13.2.0-7ubuntu1) ...  
Setting up g++-13 (13.3.0-6ubuntu2-24.04) ...  
Setting up g++ (4:13.2.0-7ubuntu1) ...  
update-alternatives: using /usr/bin/g++ to provide /usr/bin/c++ (c++) in auto mo  
de  
Setting up build-essential (12.10ubuntu1) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...  
new-user@new-user-VirtualBox: ~
```






```
new-user@new-user-VirtualBox: ~/os_lab_24k_0559
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ pwd
/home/new-user/os_lab_24k_0559
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ nano intro_24k0559.txt
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ cat intro_24k0559.txt
My Name is Bazil uddin khan
My Rollno is 24k0559
I am learning os lab
This course is interesting
I love programming
I am enhancing my problem solvig skills
I want to become a software developer
I am enhanving my front end skills
I want to become a software developer
This is programming
The Best course
The challenging
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ tac intro_24k0559.txt
The challenging
The Best course
This is programming
I want to become a software developer
I am enhanving my front end skills
I want to become a software developer
I am enhancing my problem solvig skills
I love programming
This course is interesting
I am learning os lab
My Rollno is 24k0559
My Name is Bazil uddin khan
new-user@new-user-VirtualBox:~/os_lab_24k_0559$
```

```
new-user@new-user-VirtualBox: ~/os_lab_24k_0559
My Name is Bazil uddin khan
My Rollno is 24k0559
I am learning os lab
This course is interesting
I love programming
I am enhancing my problem solvig skills
I want to become a software developer
I am enhanving my front end skills
I want to become a software developer
This is programming
The Best course
The challenging
intro_24k0559.txt (END)
```

```
Feb 1 09:22
new-user@new-user-VirtualBox: ~/os_lab_24k_0559

I want to become a software developer
This is programming
The Best course
The challenging

new-user@new-user-VirtualBox:~/os_lab_24k_0559$ tac intro_24k0559.txt

The challenging
The Best course
This is programming
I want to become a software developer
I am enhanving my front end skills
I want to become a software developer
I am enhancing my problem solvig skills
I love programming
This course is interesting
I am learning os lab
My Rollno is 24k0559
My Name is Bazil uddin khan
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ less intro_24k0559.txt
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ head intro_24k0559.txt
My Name is Bazil uddin khan
My Rollno is 24k0559
I am learning os lab
This course is interesting
I love programming
I am enhancing my problem solvig skills
I want to become a software developer
I am enhanving my front end skills
I want to become a software developer
This is programming
new-user@new-user-VirtualBox:~/os_lab_24k_0559$
```

```
Feb 1 09:22
new-user@new-user-VirtualBox: ~/os_lab_24k_0559

I am enhanving my front end skills
I want to become a software developer
I am enhancing my problem solvig skills
I love programming
This course is interesting
I am learning os lab
My Rollno is 24k0559
My Name is Bazil uddin khan
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ less intro_24k0559.txt
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ head intro_24k0559.txt
My Name is Bazil uddin khan
My Rollno is 24k0559
I am learning os lab
This course is interesting
I love programming
I am enhancing my problem solvig skills
I want to become a software developer
I am enhanving my front end skills
I want to become a software developer
This is programming
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ tail intro_24k0559.txt
This course is interesting
I love programming
I am enhancing my problem solvig skills
I want to become a software developer
I am enhanving my front end skills
I want to become a software developer
This is programming
The Best course
The challenging
new-user@new-user-VirtualBox:~/os_lab_24k_0559$
```

```
Feb 1 09:24
new-user@new-user-VirtualBox: ~/os_lab_24k_0559

I want to become a software developer
I an enhanving my front end skills
I want to become a software developer
This is programming
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ tail intro_24k0559.txt
This course is interesting
I love programming
I an enhancing my problem solvig skills
I want to become a software developer
I an enhanving my front end skills
I want to become a software developer
This is programming
The Best course
The challenging
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ echo "Additional line added in task">>intro_24k0559.txt
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ cat intro_24k0559.txt
My Name is Bazil uddin khan
My Rollno is 24k0559
I an learning os lab
This course is interesting
I love programming
I an enhancing my problem solvig skills
I want to become a software developer
I an enhanving my front end skills
I want to become a software developer
This is programming
The Best course
The challenging
Additional line added in task
new-user@new-user-VirtualBox:~/os_lab_24k_0559$
```

```
Feb 1 09:25
new-user@new-user-VirtualBox: ~/os_lab_24k_0559

This is programming
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ tail intro_24k0559.txt
This course is interesting
I love programming
I am enhancing my problem solvig skills
I want to become a software developer
I an enhanving my front end skills
I want to become a software developer
This is programming
The Best course
The challenging
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ echo "Additional line added in task">>intro_24k0559.txt
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ cat intro_24k0559.txt
My Name is Bazil uddin khan
My Rollno is 24k0559
I am learning os lab
This course is interesting
I love programming
I an enhancing my problem solvig skills
I want to become a software developer
I an enhanving my front end skills
I want to become a software developer
This is programming
The Best course
The challenging
Additional line added in task
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ echo "The file is overwritten os lab">>intro_24k0559.txt
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ cat intro_24k0559.txt
The file is overwritten os lab
new-user@new-user-VirtualBox:~/os_lab_24k_0559$
```

```
Feb 1 10:06
new-user@new-user-VirtualBox: ~/os_lab_24k_0559

new-user@new-user-VirtualBox:~/os_lab_24k_0559$ nano info_24k_0559.c
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ gcc info_24k_0559.c -o info_24k_0559
/usr/bin/ld: /usr/lib/gcc/x86_64-linux-gnu/13/../../../../x86_64-linux-gnu/Scrt1.o: in function '_start':
collect2: error: ld returned 1 exit status
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ nano info_24k_0559.c
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ cat info_24k_0559.c
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ nano info_24k_0559.c
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ cat info_24k_0559.c
cat: info_24k_0559.c: No such file or directory
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ cat info_24k_0559.c
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ nano info_24k_0559.c
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ cat info_24k_0559.c
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ ^M
: command not found
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ cd ~/os_lab_24k_0559
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ nano info_24k_0559.c
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ nano info_24k_0559.c
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ cat info_24k_0559.c
#include <stdio.h>

int main()
{
    printf("My Name is Bazil Uddin Khan\n");
    printf("Roll No:24k-0559");
    return 0;
}

new-user@new-user-VirtualBox:~/os_lab_24k_0559$ gcc info_24k_0559.c -o info_24k_0559
new-user@new-user-VirtualBox:~/os_lab_24k_0559$ ./info_24k_0559
My Name is Bazil Uddin Khan
Roll No:24k-0559new-user@new-user-VirtualBox:~/os_lab_24k_0559$
```

```
Feb 1 10:07
info_24k_0559.c
~/os_lab_24k_0559

intro_24k0559.txt
info_24k_0559.c

#include <stdio.h>

int main()
{
    printf("My Name is Bazil Uddin Khan\n");
    printf("Roll No:24k-0559");
    return 0;
}
```